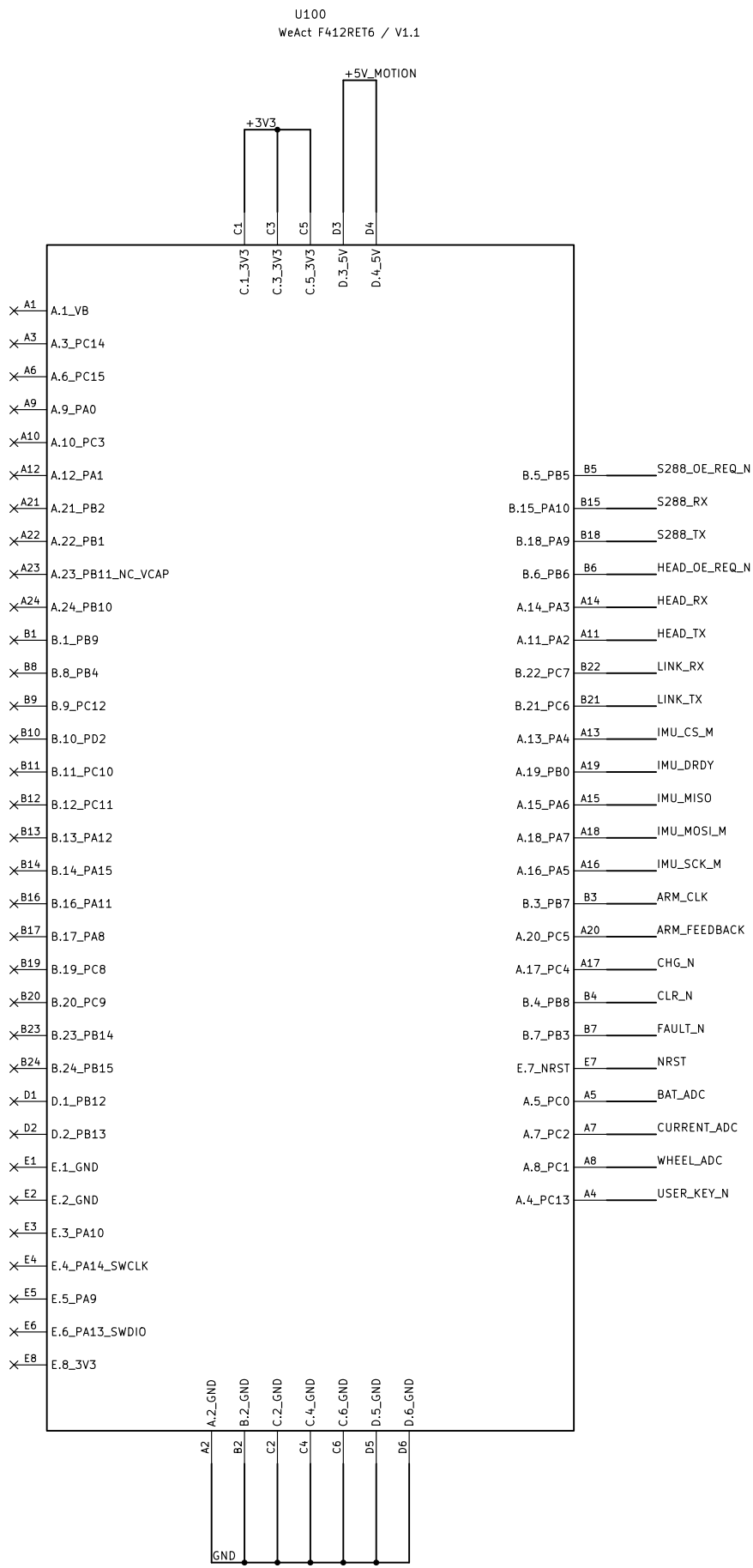
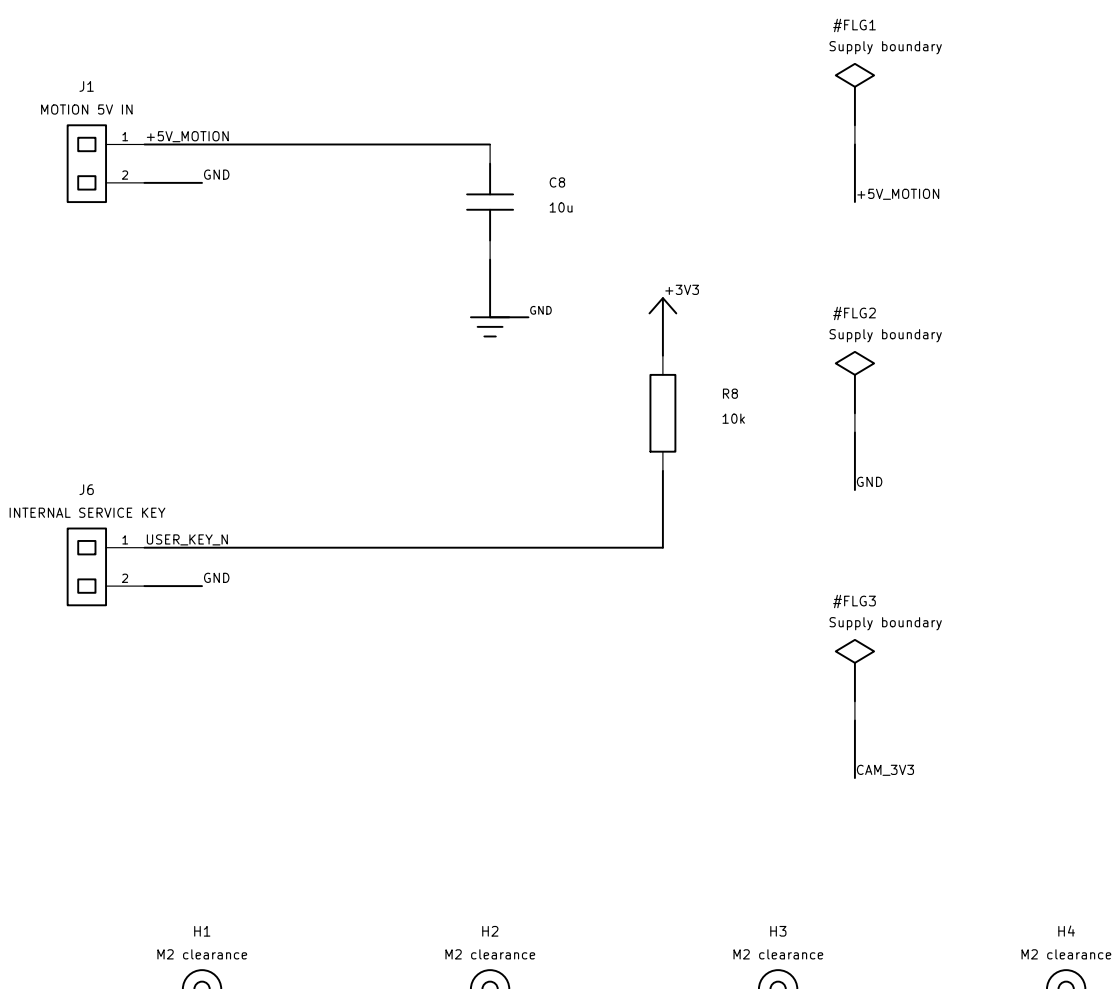


01 STM32 module / GPIO map



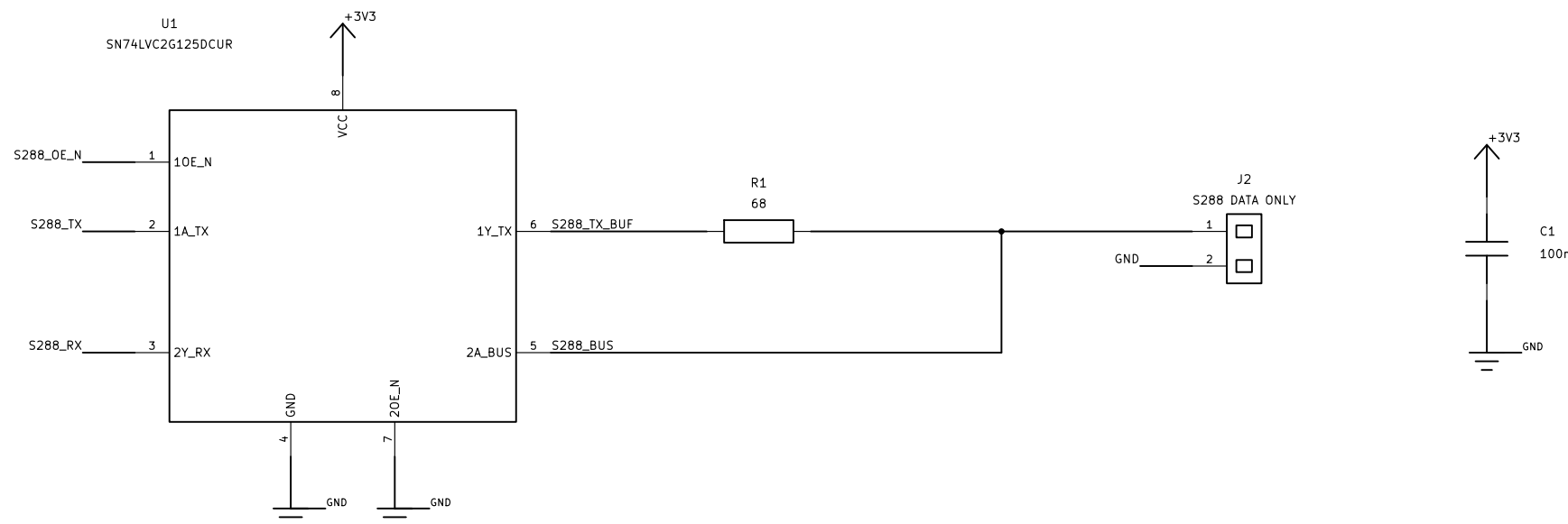
NC pins are intentionally unused. SWD probe connects on the module top.

09 Input / user button / supply boundaries



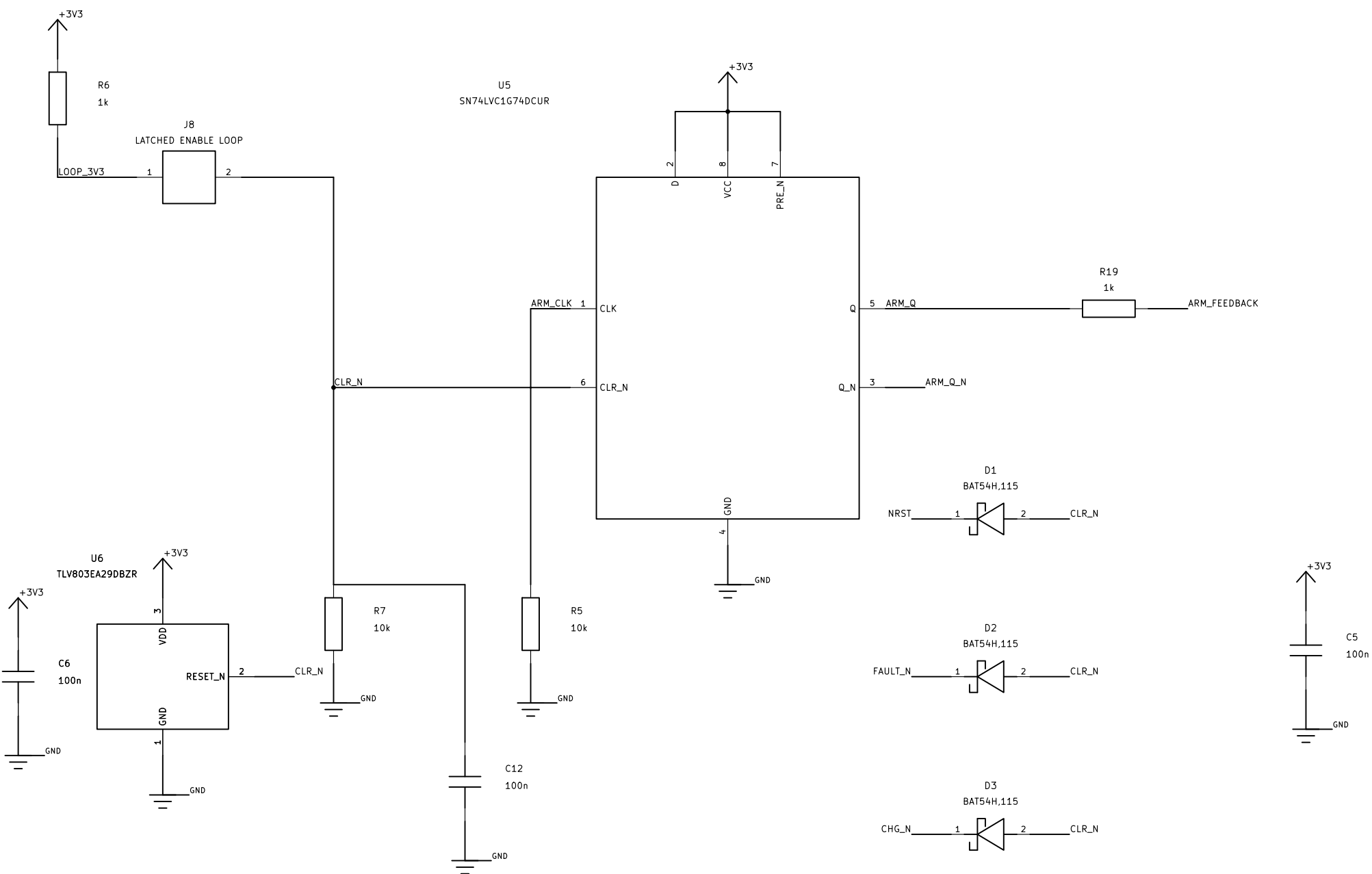
J6 is a user button. Physical disable is J6: reset can cause a fail.

02 Wheel TTL / 6 Mbps target



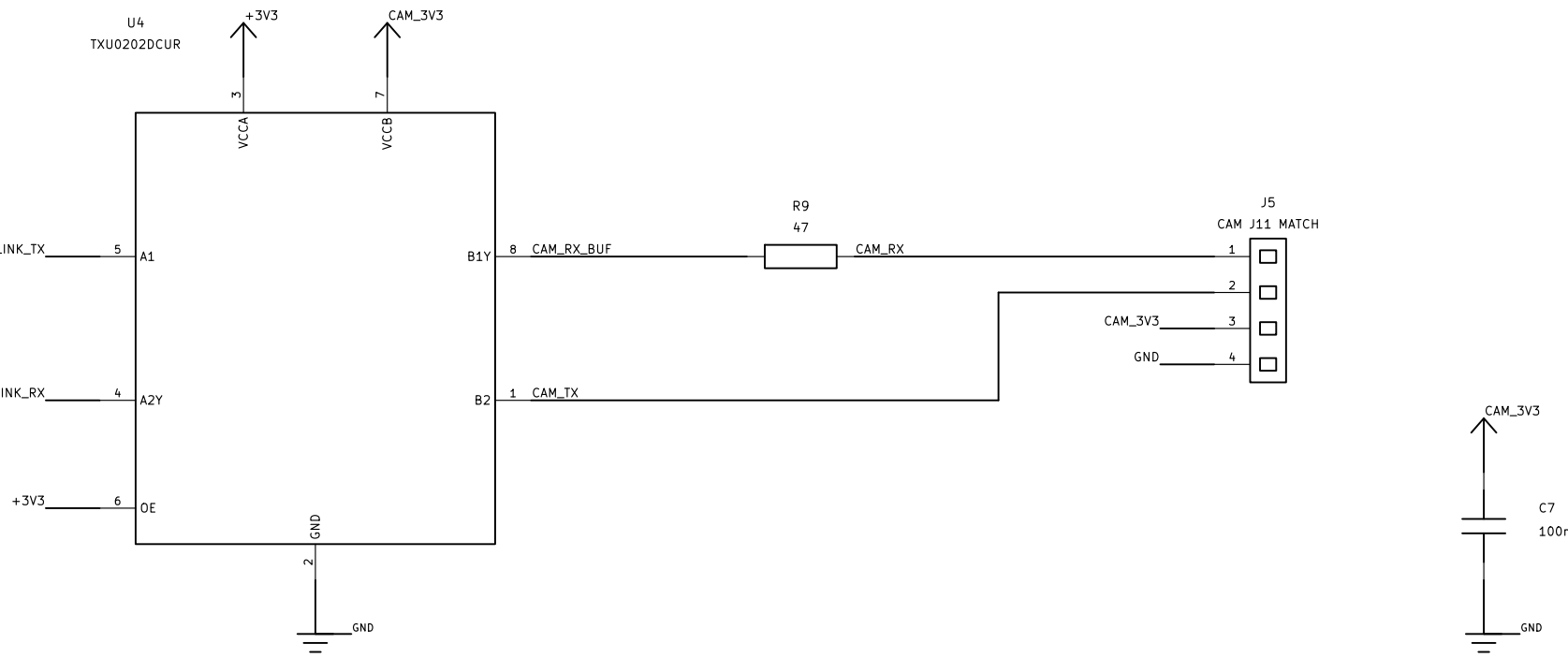
3.3 V logic; actuator thresholds and timing remain bench items.

04 Explicit ARM latch / physical disable loop



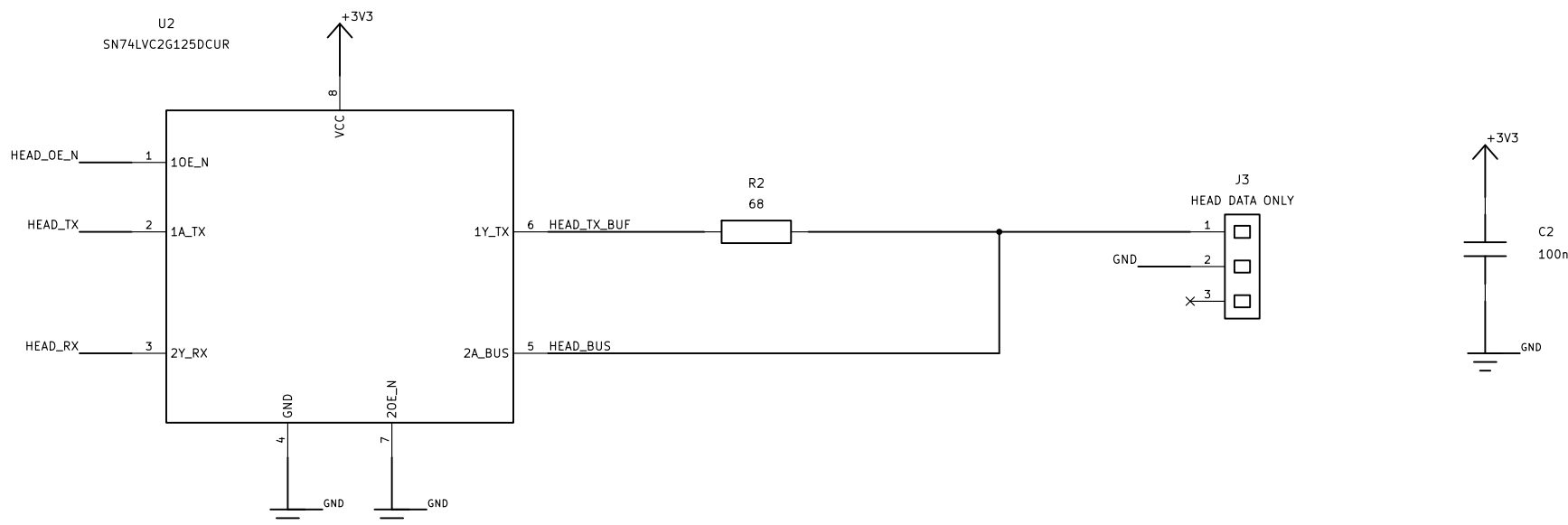
CLR_N dominates. Re-arm requires a new ARM_CLK edge; this is not a watchdog.

07 Interaction UART / power-domain isolation



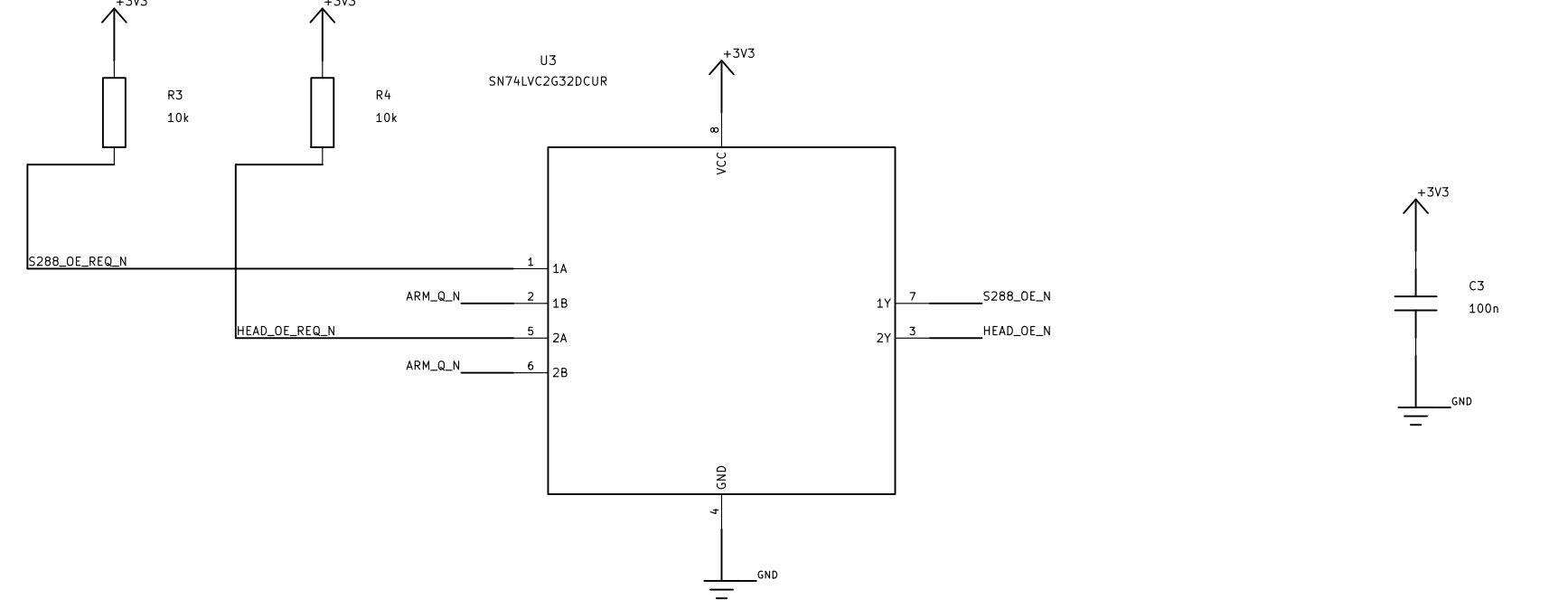
CAM_3V3 powers translator B only. It does not power the motion MCU.

03 Head TTL / separate bus



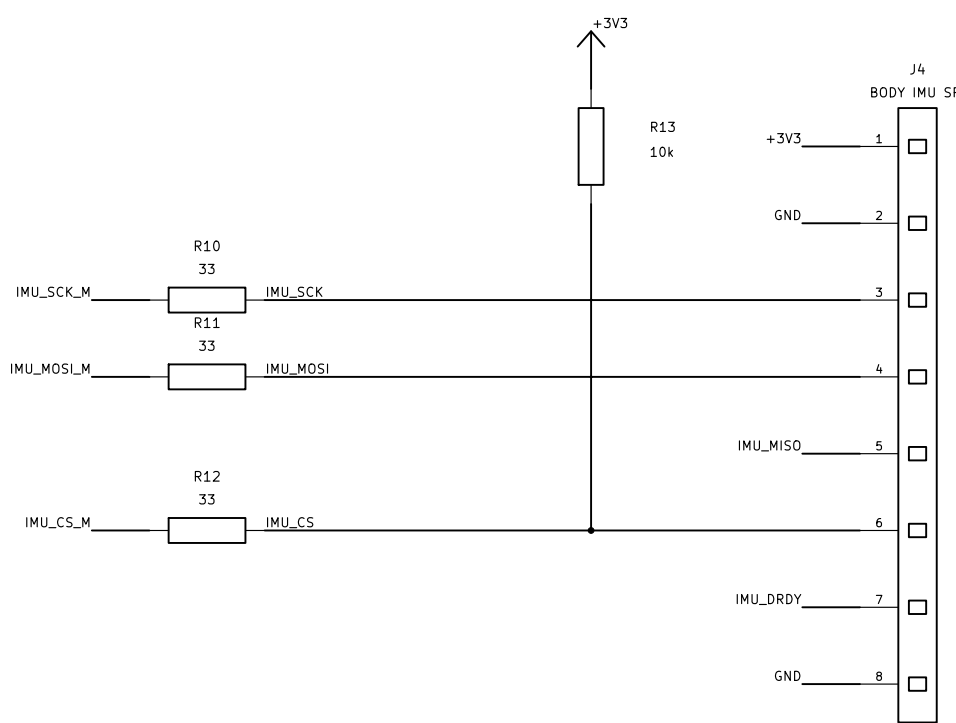
3.3 V logic; actuator thresholds and timing remain bench items.

05 Transmit-enable gating



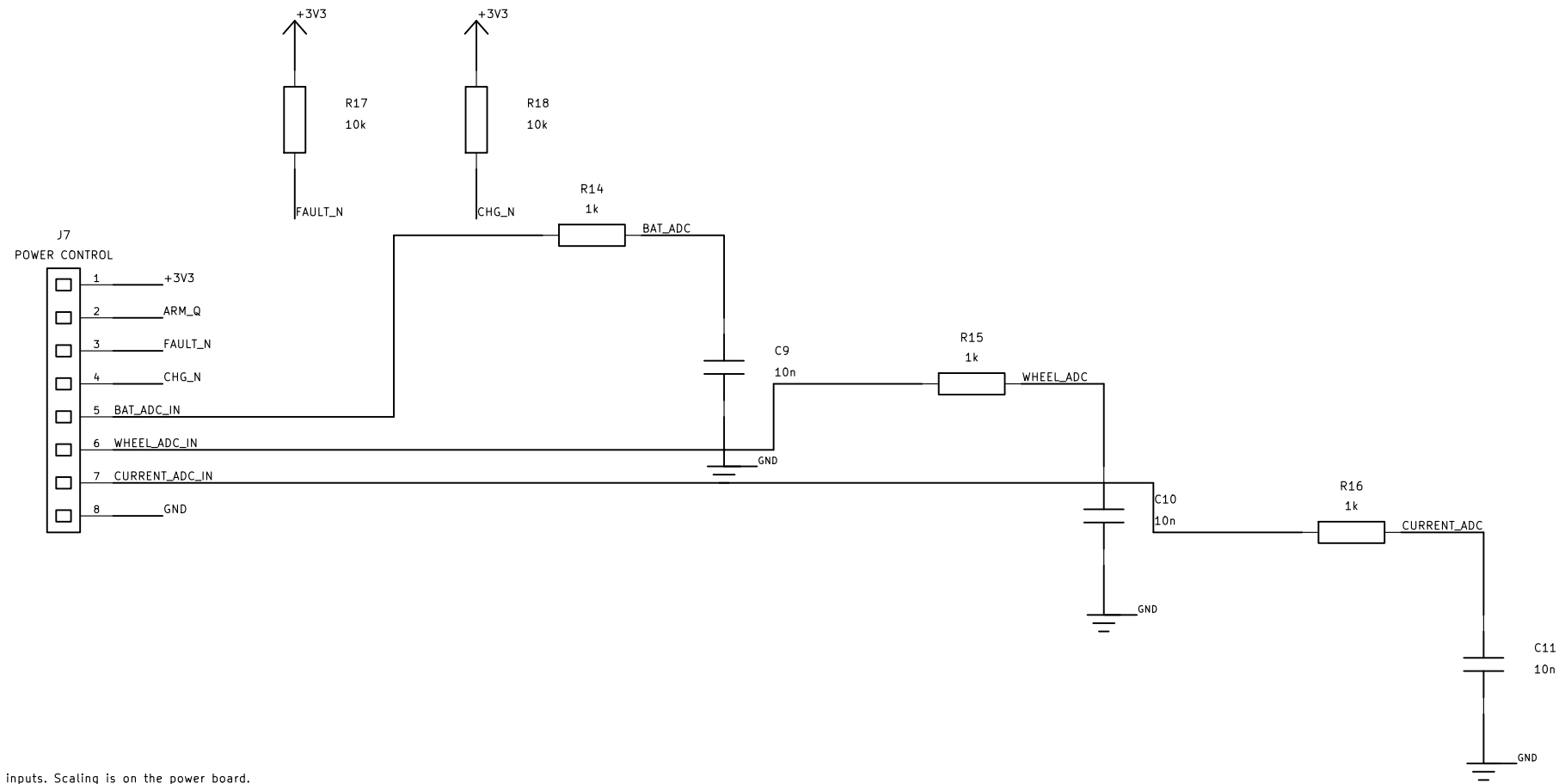
DE_N = request_N OR ARM_Q_N. Pull-ups disable TX by default.

06 Body IMU connector / source damping



SPI and DRDY belong to the rigid body IMU, not the head.

08 Power monitor / RC filters



1k / 10nF filters at the ADC inputs. Scaling is on the power board.